

Module 1

Introduction to Software Engineering & Process
Models

Introduction

- Computer software is the product that software engineers design and build.
- It consists of
 - Programs
 - Documents
 - Data (Operating procedures)
- From software engineer's point of view
 - Work product is the programs, documents and data
- From user's point of view
 - Work product is the resultant information that make user's world better.

Role of Software

- Perform two roles
 - Product → delivers computing potential surrounded by group of computers
 - Vehicle/Platform → used to deliver the product, acts as basis for OS, networks and other tools.
- Functions of software
 - Delivers product(information).
 - Provides gateway for WWW information.
 - Provides the means for sharing data.
 - Manages business information.

Program v/s software product

- Programs
 - Developed by individuals for their personal use.
 - Small in size.
 - Lacks good UI
 - Lacks proper documentation
- Software product
 - Multiple users
 - Good UI
 - Proper documentation
 - Carefully designed, implemented and thoroughly tested.

Software Engineering

- **Software** →
 - A set of instructions (programs)
 - Data Structures to manipulate information
 - Document that describes operation and use of programs.
- **Engineering** → the application of principles to develop a structure, machine, system and process.
- **Software Engineering** → It is the application of a systematic, disciplined and quantifiable approach to the development, operation and maintenance of software.
- Software Engineering is a discipline whose aim is the production of quality software.

Software characteristics/Nature of Software

- **Hardware**→ When hardware is built, the human creative process (analysis, design, construction, testing) is ultimately translated into a physical form.
- **Software**→ is logical rather than a physical system element.
- Therefore, software has characteristics that are considerably different than those of hardware:

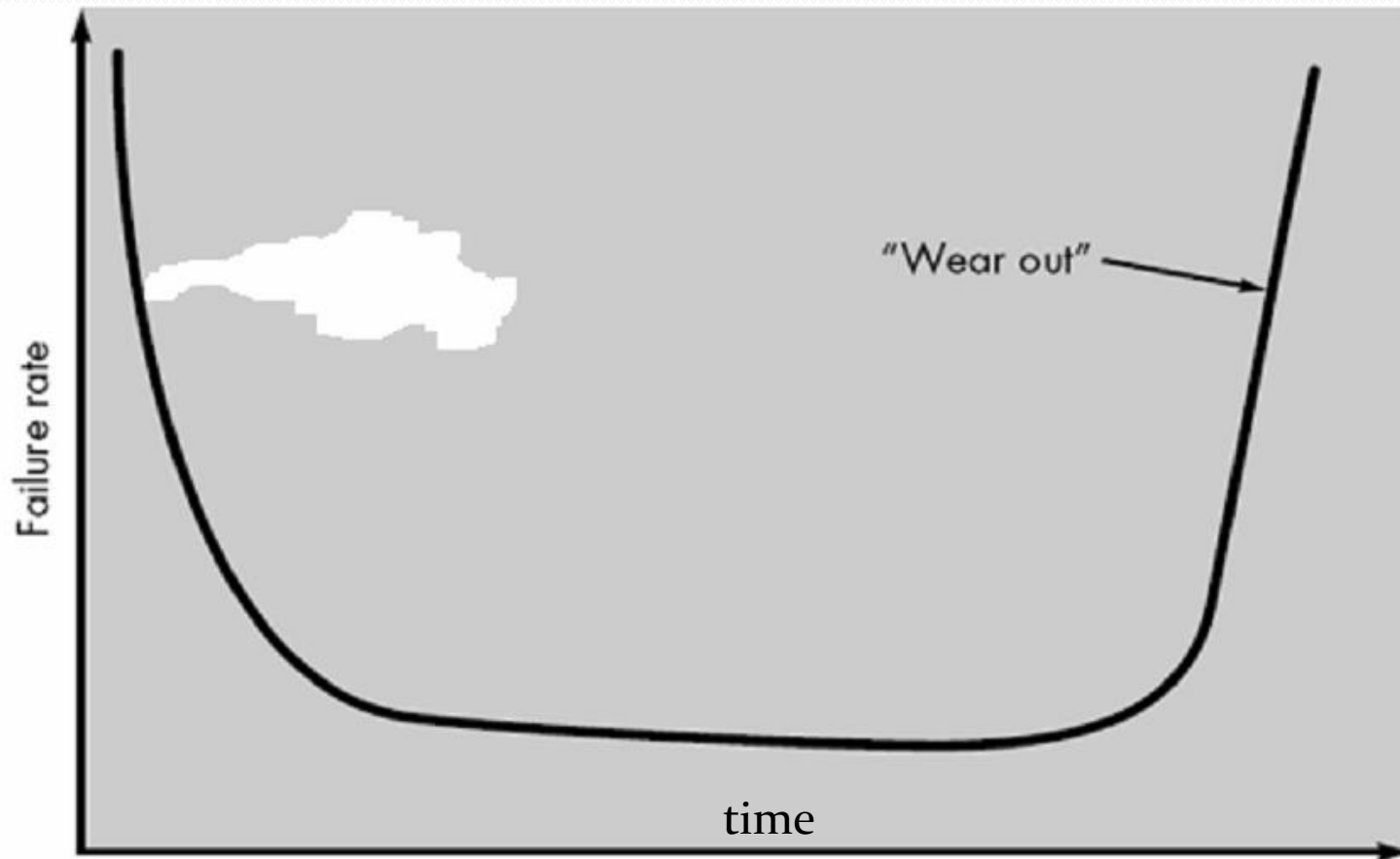
Software characteristics

- **Software is developed or engineered; it is not manufactured**
- Both activities require the construction of a "product" but the approaches are different.

Software characteristics

- **Software doesn't “wear out”**
- **Hardware**
 - Early in it's life cycle the failure rate of hardware is high
 - Defects are corrected and failure rate drops to a steady state level for some period of time.
 - As time passes- failure rate rises again as hardware components suffer from environmental hazards like dust. Temperature extremes etc.
 - Hence hardware wears out with time.

Software characteristics (Failure rate curve for H/W)



Software characteristics

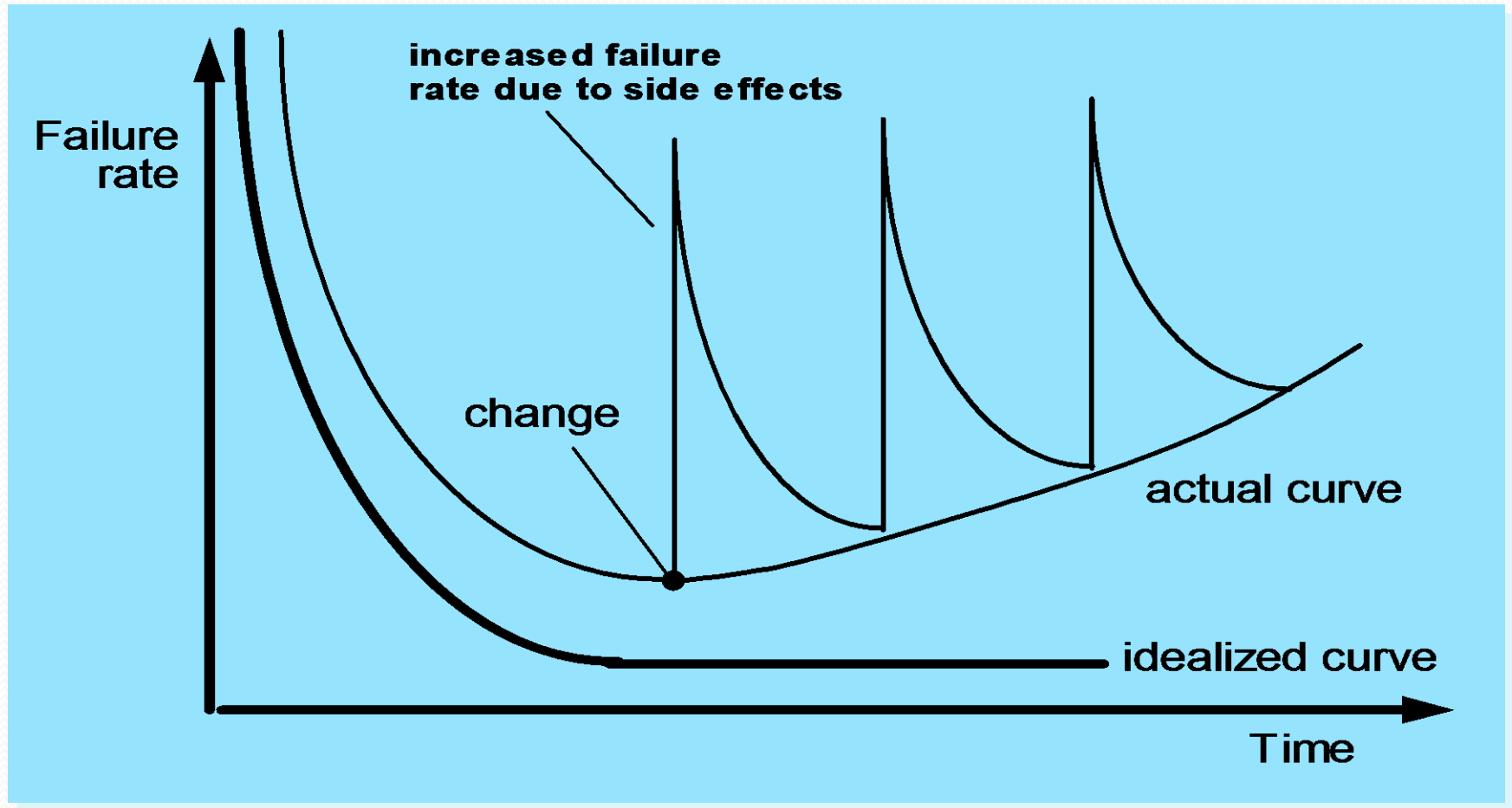
- Software
- Idealized failure rate curve for software:
 - Early life of a software- Failure rate high due to undiscovered defects.
 - Once defects are corrected the curve flattens.

Software characteristics

- Actual failure rate curve for software:
 - Software is not susceptible to environment maladies.
 - But during its life, software undergoes change(maintenance).
 - Hence **“software doesn’t wear out but deteriorates”**.
 - As a change is made new defects are introduced and the failure rate spikes.
 - Before the curve can return to it’s original steady state, another change is requested, causing the curve to spike again.
 - Slowly the minimum failure rate begins to rise.
 - Software starts deteriorating due to change

Software characteristics

(Failure rate curve for S/W)



Software characteristics

- **Although the industry is moving toward component-based assembly, most software continues to be custom built.**
- **Software is intangible**
 - Hard to understand , development effort
- **Software is easy to reproduce**
 - Cost is in its *development*.
 - in other engineering products, manufacturing is the costly stage
- **The industry is labor-intensive**
 - Hard to automate

Software Applications

- Remember as “BEEP WARS O”
- Business Software
- Engineering & Scientific software
- Embedded software
- Personal computer software
- Web based software (WebApps)
- AI software
- Real time software
- System software
- Open Source software

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- END OF TOPIC
 - NEXT TOPIC –PROCESS MODELS